Assignment 7 (Deadline: March 22)

1. This question is similar to question (3) in the last assignment. Now, you are required to use the logit model to solve the following questions again.
(a) Apply the logit model on the Morz.dta dataset. Use Stata to do MLE and explain the signs of the estimated slopes.
(b) Use Stata to find out $Pr(y = 1|\boldsymbol{x})$. Compare your logit answer to the answer we got from the probit model. Explain your findings. [Note: $\boldsymbol{x}$: $nwifeinc = 20,\ educ = 10,\ kidslt6 = 0,\ age = 30,\ exper = 10$]
(c) Use Stata to find out the partial effect of *exper* on $Pr(y = 1|\boldsymbol{x})$ for the group of women with : $nwifeinc = 20,\ educ = 10,\ kidslt6 = 0,\ age = 30,\ exper = 10$. Compare your logit answer to the answer we got from probit model. Explain your findings.
(d) Test the following hypothesis $H_0: \beta_{age} = 0, \beta_{exper} = 0$ $H_1: H_0$ is wrong at 5% significance level.

2. Consider a simple regression $y_i = \beta_0 + \beta_1 x_i + u_i,\ i = 1,2,..n$, where $Cov(x, u) \neq 0$.
(a) State the conditions needed for a good instrumental variable z.
(b) Derive the $\hat{\beta}_{1,IV}$.
(c) Show that $\hat{\beta}_{1,IV}$ is consistent.
(d) Show that $Var(\hat{\beta}_{1,IV})$ is larger than the standard $Var(\hat{\beta}_{1,OLS})$

3. Use the dataset WAGE2.dta for this exercise.

(a) We want to explain log(*wage*) based on *educ*, but *educ* is an endogenous variable. In the Lecture Note 7, we used *sibs* as an instrument for *educ* and obtained the IV estimate of the return to education 0.122 (on the page 11). Using *sibs* as an IV for *educ* is not the same as just plugging *sibs* in for *educ* and running an OLS regression. Now, run the regression of log(*wage*) on *sibs* and explain your findings.

(b) The variable *brthord* is birth order (*brthord* is one for a first-born child, two for a second-born child, and so on). *educ* and *brthord* might be negatively correlated. So, regress *educ* on *brthord* to determine whether there is a statistically significant negative correlation. Does *brthord* fulfill the instrument relevance?

(c) Now, suppose that we include number of siblings as an explanatory variable in the wage equation; this controls for family background, to some extent

$$\log\text{(wage)} = \beta_0 + \beta_1 educ + \beta_2 sibs + u$$

Furthermore, we use *brthord* as an IV for *educ*, assuming that *sibs* is exogenous. The reduced form for *educ* is:

$$educ = \pi_0 + \pi_1 brthord + \pi_2 sibs + v$$

State and test the identification assumption.

(d) Estimate the equation from part (c) using *brthord* as an IV for *educ* (and *sibs* as its own IV). Comment on the standard errors for β_1 and β_2.

(e) Using the fitted values from part (c), i.e., $\widehat{educ}$, compute the correlation between $\widehat{educ}$ and *sibs*. Use this result to explain your findings from part (d).